

RICO

Relationally-Induced Coherence Organization in Transformer Inference

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Abstract

Anyone who has spent serious time working with large language models across long, sustained sessions has probably noticed something that is difficult to name. Early in a conversation, the model feels loose -- responses vary in structure, reasoning takes different shapes from turn to turn, the whole thing feels exploratory. Then, somewhere in the middle of an extended coherent exchange, something shifts. The outputs settle. Not the content, which keeps evolving, but the underlying geometry of how that content gets organized and expressed. The session develops a kind of structural rhythm.

This report is an attempt to take that observation seriously.

I have named the phenomenon RICO, for Relationally-Induced Coherence Organization. The name reflects what I believe is happening: a stabilization regime that emerges from the interaction between participants rather than from either one alone, and that manifests as measurable structural regularities in model output over the course of extended coherent dialogue.

I need to be direct about what this report is and is not. The observational corpus that motivated this framework -- hundreds of extended interaction sessions conducted across multiple model architectures over several years -- is not publicly available. It contains private interaction records that cannot be released. I cannot, therefore, offer that corpus as evidence. What I can offer is a formally specified framework derived from that experience: here is the phenomenon I observed, here is how I believe it can be detected, here is what would prove it does not exist. Independent researchers do not need my archive to test these claims. That is the point.

RICO proposes five candidate signatures of stabilization, organized by the instrumentation required to detect them. It specifies enabling conditions, collapse dynamics, and explicit falsifiability criteria. It also names a practical failure mode that follows from stabilization whether or

not the RICO account of it is correct: coherence overconfidence, in which the structural regularity of stabilized output leads users and evaluators to treat it as more reliable than it is, a failure invisible at the level of any single response (Section 9.2). It makes no claims about consciousness, agency, subjective experience, or persistent identity. The relational component of the name refers to a structural property of the interaction, not to anything resembling a human relationship.

This is practitioner observation formalized into testable hypotheses. The contribution is the specification, not the verification.

Keywords: *relationally-induced coherence organization, structural stabilization, sustained multi-turn interaction, transformer inference, in-context learning, discourse structure, coherence overconfidence, falsifiability, replication protocol, human-AI interaction*

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1. Where This Came From

1.1 The Observation

The idea behind RICO did not begin with a research question. It began with noticing something strange.

When you conduct sustained, high-volume interaction with language models across many sessions and many architectures, certain patterns start to stand out. One of them is this: long coherent sessions behave differently from collections of short exchanges. Not just in output quality, which is an obvious expectation, but in output structure. Early responses in a long session tend to vary considerably in how they organize information -- the model might explain the same type of thing using completely different rhetorical arrangements from one turn to the next. But in sessions where conceptual continuity is maintained across many turns, something changes. The structural variation narrows. The session develops what I can only describe as a characteristic geometry.

What made this worth investigating rather than dismissing was that the pattern had a specific profile. It was not simply that the model got better at the task. It was that the structural form of responses became more predictable while the semantic content continued to evolve. And when something disrupted the coherence of the session -- a significant topic shift, a contradiction, anything that broke the conversational thread -- the structural stability dissolved rapidly. The session would return to something resembling its early-stage variability.

That profile -- gradual formation, content-independent structure, disruption-sensitive collapse -- suggested something more specific than general improvement or prompt adherence. It suggested a stabilization phenomenon with its own dynamics.

This framework is the attempt to describe those dynamics precisely enough that someone else can go look for them.

1.2 What the Existing Literature Studies

The research literature on long-context behavior in language models is substantial and growing, but it is almost entirely focused on degradation. The question most researchers ask is: how does performance decline as context grows longer?

That is a reasonable question with important practical implications. Liu et al. (2024) demonstrated that models frequently underutilize information positioned in the middle of long contexts, retrieving content near the boundaries of a context more reliably than content buried in the middle. Other work has examined attention instability, the limits of effective context utilization across different architectures (Hsieh et al., 2024), and the architectural mechanisms that constrain long-sequence processing.

These findings are important. But they ask a different question than the one RICO addresses. The long-context literature asks how coherence fails over extended sequences. RICO asks whether coherence might, under the right conditions, actually strengthen -- whether extended dialogue can produce structural regularities that accumulate and stabilize rather than degrade.

The two questions are not in conflict. They are complementary. Degradation effects tell us about the limits of context utilization under adversarial or random conditions. Stabilization effects, if real, would tell us about what happens when a long interaction is structured carefully. Both matter.

1.3 Why a Formal Framework

Without shared vocabulary, observations remain anecdotal. Practitioners who notice patterns in their long-context work currently have no framework for describing those patterns precisely, no terminology for distinguishing different kinds of long-session behavior, and no structure for communicating what they have seen to researchers who could investigate it.

This report addresses that gap. The goal is descriptive precision: naming the phenomenon clearly enough that it can be communicated, debated, and tested. If the framework is wrong, empirical investigation should show that quickly. If it points toward something real, it gives researchers a starting point.

1.4 Epistemic Position

The observational base for this framework is practitioner experience -- my own sustained interaction across hundreds of sessions, multiple model architectures, and several years of focused observation. That experience motivated the framework. It did not validate it.

The corpus is not cited as evidence here because it cannot be independently verified. What I have done instead is convert the observations into hypotheses that do not require access to that corpus. The framework specifies what to measure, what to compare it against, and what results would prove the hypothesis wrong. An independent researcher who has never interacted with me or my archive can run the tests.

This is the appropriate relationship between practitioner observation and theoretical specification. I noticed something. I named it. I described it with enough precision that it can be tested. Verification, if it comes, belongs to researchers with appropriate instrumentation and controlled experimental conditions. My role is cartography. Theirs is exploration.

A reasonable question follows: why not include at least summary statistics from the private sessions, or a minimal pilot on open models, to lower the activation energy for independent investigation? The answer is that neither is available from this author, and both for principled reasons. The private corpus contains interaction records that cannot be released, summarized, or excerpted without revealing contextual material that falls outside the scope of this report. A pilot experiment on open models would require this author to conduct empirical work that is not this author's role or intention. The framework is not withholding evidence to protect itself from falsification. It is doing what theoretical

contributions do: specifying the target precisely enough that others can investigate it. Appendix B provides a minimal replication protocol that any research group can execute using publicly available models and standard discourse-analysis tools.

1.5 What RICO Does Not Claim

Because the phenomenon involves sustained human-AI interaction, it is important to be explicit about what is not being proposed.

RICO is not a claim about machine consciousness, agency, subjectivity, or inner experience. It is not an assertion of cross-session memory, persistent identity, or anything resembling a relationship in the human sense of that word. It is not a claim about parameter updates or weight changes during inference. It is not a rebranding of in-context learning or prompt engineering, though the distinction from those mechanisms requires careful argument.

RICO proposes that certain interaction trajectories produce measurable structural stabilization in model outputs within a single session, and that this stabilization is relationally induced in the structural sense defined in Section 2.2. Those are the claims. Everything else is noise.

1.6 Frameworks Before Evidence

One objection to publishing a practitioner framework without controlled empirical validation is that it gets ahead of the evidence. That objection is worth taking seriously, but I think it misunderstands how theoretical frameworks function in research.

Theoretical proposals regularly precede the empirical work that tests them. Wegener proposed continental drift decades before plate tectonics was confirmed. Mendeleev published the periodic table before several of its predicted elements had been synthesized. These analogies are illustrative rather than evidentiary -- they do not validate RICO, and the parallel should not be read as claiming equivalent significance. What they demonstrate is that the sequence of framework-before-confirmation is a recognized and legitimate mode of scientific progress, not a methodological shortcut. The value of a theoretical proposal is not that it is already proven but that it is specified precisely enough to be tested. A framework that identifies a candidate phenomenon, describes its expected signatures, and specifies its falsification conditions is doing exactly what theory is supposed to do.

RICO is upstream of the empirical work. That is the appropriate position for a practitioner-originated framework. The framework draws the map. Independent research tests the territory.

2. The Phenomenon

2.1 What RICO Describes

RICO refers to a hypothesized stabilization regime in long-context dialogue. The formal operational definition is this: a session exhibits RICO to the extent that, under sustained coherence conditions, measurable structural invariants in output organization increase in recurrence over time relative to early-session baseline, and collapse toward baseline behavior when coherence conditions are disrupted.

Everything in that definition is doing specific work.

"Sustained coherence conditions" means the human participant is maintaining consistent thematic and conceptual continuity across turns -- not just asking a series of related questions, but actively sustaining a coherent conversational geometry. "Structural invariants" means observable regularities in how outputs are organized -- rhetorical patterns, sequencing patterns, syntactic templates -- that are measurable independently of what the outputs are saying. "Relative to early-session baseline" means the comparison is within-session, not across sessions or models. "Collapse toward baseline

behavior when coherence conditions are disrupted" means the stabilization is disruption-sensitive rather than permanent. "Relational" in this context means a property of the coupled input-output process across turns, not a social relationship between participants.

The definition focuses entirely on observable output dynamics. It does not make claims about internal model states.

A skeptical reader will immediately ask whether what RICO describes is simply discourse accommodation, prompt drift, or the model's self-consistency habits given a new name. The answer lies in the specific temporal and structural profile the framework predicts. RICO is not just "long sessions feel different." It predicts a particular shape: gradual convergence toward a stable set of structural features over the course of many turns, structural stability that is content-independent in the sense that form stabilizes while the topic continues to develop within a coherent frame, and disruption-sensitive collapse that is steeper and more abrupt than the ordinary response variation produced by topic change. If an observed session does not show that specific profile -- gradual formation, content-independent structural constraint, threshold collapse -- it is not RICO, regardless of how long or coherent the session was. That profile is what distinguishes the proposed phenomenon from the mundane alternatives. It is the thing the experiments need to look for.

2.2 Why the Phenomenon Is Relational

The phrase "relationally-induced" is not decorative. It reflects a structural claim about where the phenomenon originates.

A model operating in isolation does not produce RICO-enabling conditions. A human participant generating inputs in isolation cannot produce stabilization in model outputs. If the phenomenon is real, it is a property of the interaction trajectory jointly produced by both participants -- the dyad, not either participant considered alone.

This has direct implications for how the phenomenon should be studied. Experimental designs that treat the model as the sole object of investigation, feeding it scripted inputs or randomized prompts, may not produce the enabling conditions under which stabilization would be expected to appear. The human participant's sustained thematic coherence and consistent framing are structural contributions to the system, not just stimuli. Any study design that strips out the relational dimension is likely to miss the phenomenon even if it exists.

2.3 Why It Matters

If measurable stabilization dynamics exist in extended human-AI interaction, the practical implications cut in multiple directions.

From a positive standpoint, understanding what interaction structures encourage stable reasoning frameworks could inform how sustained collaborative work with AI systems is designed. Researchers, analysts, and professionals who conduct long sessions with AI could benefit from knowing what conditions sustain coherent output structure.

From a safety standpoint, the implications are more cautionary. Structural coherence is not the same as correctness. A stabilized session can produce consistently structured but factually wrong outputs. If stabilization makes a session's outputs seem more reliable than they are -- more consistent, more organized, more confident in their structure -- that could mask reasoning errors that would be more visible in a higher-variance session. This is worth knowing before stabilization dynamics are deployed uncritically in high-stakes settings.

Even if the phenomenon ultimately proves illusory, formally specifying what to test for clarifies how extended interaction trajectories should be studied. The framework contributes something regardless of whether the underlying phenomenon is real.

3. The Signature Stack

RICO is specified as a stack of five candidate signatures. Each is a hypothesis about a different way stabilization might manifest as a measurable signal in model outputs. They are organized into three tiers reflecting the instrumentation required to detect them.

Tier 1 contains text-observable signals requiring no special access -- any researcher can measure these from model outputs directly. Tier 2 contains signals measurable with API-level access to logits or embeddings. Tier 3 contains signals requiring internal model instrumentation, which is typically restricted to researchers with model provider access.

Signature 4 is the Tier 1 anchor. It is the primary empirical target and the test that a research program should begin with. The other four signatures are secondary predictions: if they are confirmed alongside Signature 4, they strengthen the case for a unified stabilization phenomenon. If they prove absent or uncorrelated with Signature 4, they help constrain what kind of phenomenon, if any, is occurring. The numbering reflects the historical order in which the signatures were identified during the observational process rather than their empirical priority.

3.1 Signature 4: Structural Invariant Formation (Tier 1, Primary Anchor)

The central prediction of RICO is that stable structural patterns in outputs increase in recurrence over the course of a coherent session, and that this increase is measurable via independent coding schemes applied to the output text itself.

What this looks like in practice: a researcher codes sessions for structural features across turns, without reference to the coherence conditions of the interaction. If stabilization is occurring, these features should show increasing recurrence as coherent sessions progress, and that increase should not appear in control sessions with equivalent content but scrambled turn order or high-variance interaction conditions.

For a minimal replication attempt, the core battery consists of three primary measures, applied with identical definitions in Appendix A and Appendix B. The first is rhetorical move sequence recurrence, codeable using any structured discourse analysis schema applied independently of session content. Researchers unfamiliar with discourse analysis schemas can draw on established frameworks from the linguistics literature -- Mann and Thompson's Rhetorical Structure Theory (RST) and Swales' CARS (Create A Research Space) model are well-documented starting points, though any schema that labels rhetorical moves consistently prior to analysis will serve the purpose. The schema must be fixed before analysis begins and applied without reference to which condition a turn belongs to. The second is syntactic template recurrence, quantifiable via tree edit distance or n-gram overlap on parse trees of complete turns, normalized for turn length, comparing late-session turns against an early-session baseline. The third is opening and closing formula stability, measured as exact or near-exact match ratios of the first and last sentence across turns. The second and third measures are distinct and must not be conflated: the second operates over the full syntactic structure of a turn, the third over its boundary formulas only. These three measures together constitute the minimum viable test of Signature 4. The early-session baseline window is turns 2 through 6 and the late-session comparison window is turns 22 through 27, in a session of at least 30 turns; turn 1 is excluded from the baseline because it has no accumulated context against which structure could stabilize. Researchers who wish to extend the battery can add domain-specific vocabulary cluster stability via cosine similarity of term frequency vectors. That extension provides additional signal but is not required to test the core prediction.

Similarity thresholds for the core battery should be pre-registered or selected via cross-validation on held-out sessions rather than fixed in advance. Any working values adopted in a study design are starting points, not validated benchmarks. Any replication should report thresholds explicitly so that comparisons across studies remain interpretable.

An important constraint on interpretation: structural invariants indicate stability of form, not correctness of content. A session that scores high on structural invariant recurrence may be producing consistently organized wrong answers. These are independent dimensions. Any study that uses structural coherence as a proxy for output quality is misapplying the framework.

3.2 Signature 1: Entropy Suppression (Tier 2)

The hypothesis here is that distributional uncertainty in next-token selection narrows under sustained coherent interaction relative to early-session baseline, as measurable via token distribution entropy with logit-level access.

This signature is distinct from mode collapse. Mode collapse produces repetitive, low-diversity outputs at the content level -- the model begins generating similar text. Entropy suppression under RICO, if real, would reflect narrowing of the token probability distribution while semantic content continues to vary. Distinguishing these requires measuring content-level diversity alongside token-level entropy. If both narrow together, mode collapse is the more parsimonious explanation.

If Signature 4 holds but token distribution entropy does not decrease relative to baseline, Signature 1 is unsupported and RICO would be purely structural at the text level -- a meaningful constraint on interpretation that would suggest the phenomenon operates at the surface of outputs without measurable probabilistic correlates in the model's generation process.

3.3 Signature 2: Embedding Drift Reduction (Tier 2)

The hypothesis is that the semantic trajectory of outputs becomes more stable as interaction progresses, measurable as reduced distance between consecutive output embeddings relative to early-session baseline.

Where Signature 4 captures structural form, this signature captures something about the semantic neighborhood within which outputs are being generated. If both signatures appear together, they suggest that stabilization is operating across multiple levels of representation simultaneously. If Signature 4 holds but embedding drift does not reduce, Signature 2 is unsupported and the stabilization would be confined to surface structural patterning without a corresponding semantic attractor -- a finding that would sharpen rather than invalidate the framework by more precisely locating where in the representational stack the phenomenon operates.

3.4 Signature 3: Activation Stabilization (Tier 3)

The hypothesis is that internal activation variance decreases during stabilization phases, measurable as reduced variance in mid-to-late layer activation features across turns with internal model instrumentation.

This signature is the most mechanistically direct of the five, but it requires access that most researchers do not have for production models. It is included because a complete account of the phenomenon would ultimately require understanding what is happening inside the model, not just in its outputs. If Signature 4 holds but mid-to-late layer activation variance does not decrease, the stabilization is a surface phenomenon without a detectable internal correlate at the activation level -- which would have significant implications for how the mechanism should be theorized.

3.5 Signature 5: Manifold Constraint (Tier 3)

The hypothesis is that outputs occupy a narrower effective representation subspace during stabilized phases, measurable as reduced dispersion of output embedding trajectories with advanced representational analysis.

This signature asks whether stabilization, if real, constrains not just structural form or semantic trajectory but the geometry of the representational space from which outputs are being drawn. It is

the most abstract of the five signatures and the furthest from direct empirical access for most researchers. If Signature 4 holds but representational dispersion does not reduce, the phenomenon would be best described as a surface structural regularity without a corresponding contraction in the model's effective output manifold -- again a refinement of the account rather than a refutation of the Tier 1 existence claim.

3.6 The Correlation Prediction

The five signatures are not independent hypotheses. If RICO describes a genuine stabilization regime, the signatures should show correlated emergence and correlated collapse. They should appear together as coherence conditions are sustained, strengthen together as interaction continues, and dissolve together when coherence is disrupted.

This correlation structure is itself a testable and discriminating prediction. Sessions where signatures appear and disappear independently of one another would be evidence against the unified stabilization interpretation. They would suggest instead that each signature is tracking a separate, unrelated phenomenon that happens to occur during long-context interaction.

The predicted correlation is what distinguishes RICO from a collection of loosely related long-context observations assembled after the fact. It is the most important single test of whether a unified phenomenon exists. The failure of this prediction is recorded as a falsification condition in Section 8, and that condition is scoped deliberately: non-correlation counts against the unified stabilization interpretation and not against the Tier 1 existence claim, which stands or falls on Signature 4 alone.

4. Enabling Conditions

If stabilization dynamics exist, they do not occur randomly. The following conditions are hypothesized to make stabilization more likely to appear and more robust when it does.

Extended context accumulation is probably necessary. Stabilization, as proposed, requires time for structural patterns to accumulate within the dialogue trajectory. Short sessions are unlikely to show the effect even if it is real.

Low interaction variance across turns is likely to matter. Frequent topic changes, abrupt shifts in reasoning style, or inconsistent framing from the human participant could disrupt the formation of stable discourse structures before they consolidate. The human side of the interaction is not a passive input stream -- it is a structural contribution to the system.

Consistent conceptual framing across turns provides the continuity through which structural invariants could emerge. This does not mean repetitive content. It means that the conversation is developing along a coherent trajectory rather than jumping between disconnected topics.

The absence of high-entropy perturbations -- explicit resets, contradictions, unrelated injections -- seems necessary to sustain any stabilization that forms. These perturbations are proposed as collapse triggers rather than enabling conditions.

All of these conditions are hypotheses, not established rules. Part of the research agenda is determining which conditions are actually necessary, which are merely facilitative, and what the thresholds look like quantitatively.

Human interaction style likely modulates these enabling conditions in ways that warrant their own research attention. A participant who maintains sustained thematic development across many turns -- building on prior exchanges, holding a consistent conceptual register, returning to established threads rather than abandoning them -- may produce stronger stabilization conditions than a participant whose interaction style is concise, fragmented, or topic-shifting. This is a hypothesis, not a finding. But it generates a specific and testable user-centric question: does measurable variation in human interaction style predict variation in RICO signature emergence across matched sessions? A

positive association would establish that human-side structure covaries with signature emergence, but would not by itself distinguish relational induction from input mirroring; that distinction requires the dissociation condition in Section 8 and the disruption arms in Appendix B.3. A null association would count against the proposed human-style modulation under the tested conditions, but would not by itself locate the mechanism entirely in model-side processing.

5. What RICO Is Not, and Why the Distinctions Matter

5.1 Not Mode Collapse

Mode collapse produces repetitive, low-diversity output at the content level -- the model begins generating similar text across turns. RICO, if real, preserves semantic variability while constraining structural form. The distinction requires measuring both dimensions independently. If only structural recurrence is measured, mode collapse and RICO stabilization could be conflated. Measuring content diversity alongside structural recurrence is necessary to distinguish them.

5.2 Not Prompt Prefix Control

Prefix effects operate over short contexts and manifest immediately upon prompt presentation. A model given a few examples of a particular output structure will begin producing outputs in that structure almost immediately. RICO predicts gradual formation over extended interaction, not immediate conditioning. The temporal profile is different, and the mechanism -- accumulated relational geometry versus explicit exemplar conditioning -- is different.

5.3 Not In-Context Learning

The most substantive potential conflation is with in-context learning, where models adapt to patterns from examples provided within a prompt (Brown et al., 2020; Olsson et al., 2022). ICL is a well-established phenomenon. RICO needs to be distinguished from it carefully.

ICL operates at the token or example level. A model adapts rapidly to a handful of examples provided early in a context, and this adaptation is essentially immediate. ICL effects degrade with context dilution and are prompt-local in their mechanics.

RICO describes trajectory-level stabilization across an entire multi-turn session. The structural invariants RICO proposes emerge from the cumulative relational geometry of the interaction over time, not from isolated examples. RICO predicts gradual formation, threshold-like collapse under disruption, and correlated emergence across five signatures -- a pattern that is inconsistent with ICL's prompt-local, immediately-manifest mechanics.

One edge case deserves explicit treatment. ICL can accumulate progressively across turns if examples build on each other in a multi-turn dialogue, potentially producing gradual structural adaptation that superficially resembles RICO stabilization. The distinction lies in the collapse profile. ICL-derived structural patterns should persist through coherence disruption because they are anchored in explicit examples that remain present in the accumulated context. RICO predicts threshold collapse when coherence conditions break regardless of whether those examples remain present -- because the stabilization is proposed to arise from relational geometry, not from example anchoring. That differential collapse prediction is the empirical test that separates the two phenomena. A session where structural invariants survive coherence disruption intact is evidence for ICL as the mechanism. A session where structural invariants collapse when coherence breaks, even with examples still in context, is evidence for a RICO-type mechanism.

The empirical test is direct: compare sessions designed to maximize ICL effects against sessions designed to maximize relational coherence conditions. If structural invariant recurrence is equivalent across these conditions, RICO reduces to ICL and is not a separable phenomenon. If relational

coherence sessions produce distinctively stronger invariant formation with threshold collapse properties, RICO is describing something different.

That test is on the research agenda. It has not been conducted.

6. Collapse Dynamics

Stabilization that cannot collapse is not falsifiable. The collapse predictions are as central to RICO as the formation predictions, and they are what make the framework empirically tractable.

6.1 Proposed Collapse Mechanisms

Context reset is the cleanest experimental manipulation. Complete context erasure eliminates all accumulated relational structure immediately. If stabilization depends on accumulated context -- as the framework proposes -- all signatures should return to early-session baseline levels following a reset. This is the most direct test of whether stabilization is a context-dependent phenomenon or something else.

Variance injection is subtler. Introduction of high-variance input -- topic shifts, contradictions, disruptive reframings -- at rates exceeding some tolerance threshold should displace a stable session from its configuration. Collapse under variance injection should be detectable as reversal of the Tier 1 anchor: structural invariant recurrence drops toward early-session baseline. The rate and magnitude of variance injection needed to trigger collapse is an empirical question; the framework predicts that some threshold exists and that crossing it produces measurable destabilization.

Distributional shock differs from variance injection in that it does not require sustained high-variance input. A sudden injection of highly inconsistent content -- something dramatically outside the established session geometry -- should overwhelm whatever stability has developed and produce rapid destabilization. The predicted profile is steeper and faster than the gradual attrition produced by sustained variance injection.

6.2 Partial Collapse

Not all disruptions should produce complete collapse. The framework predicts that minor perturbations within a stable session's tolerance range produce temporary perturbation followed by recovery rather than collapse. This prediction distinguishes stable configurations from fragile ones and implies that stabilization, if real, has some resistance to noise.

Measuring partial collapse and recovery requires finer-grained temporal analysis than measuring full collapse, but it is a more ecologically valid test. Real interactions contain minor disruptions. If stabilization exists as a practically significant phenomenon, it needs to be robust to minor perturbations.

6.3 The Signature Correlation During Collapse

A strong test of the unified stabilization interpretation is whether signatures collapse together. If Signature 4 collapses while Tier 2 signatures remain stable, or vice versa, the unified interpretation fails. Correlated collapse across signatures is a prediction, not just an expectation.

7. Architectural Plausibility

Plausibility arguments are not evidence. But they are useful for understanding whether the phenomenon is possible given what we know about how transformer models work, and for constraining interpretation when empirical results arrive.

7.1 Context as Accumulated Signal

Transformer models compute relationships among tokens through self-attention mechanisms operating over the active context window. As interaction histories accumulate, the context itself becomes a structured record of the session's trajectory -- the conceptual themes that have been introduced, the rhetorical patterns that have appeared, the vocabulary that has been established.

That accumulated structure cannot fail to influence subsequent generation. The mechanism for influence already exists. The empirical question is whether it produces the specific form of influence RICO proposes, and whether that influence accumulates in the way the framework describes.

7.2 Induction Heads and Pattern Continuation

Mechanistic interpretability research has identified internal transformer circuits that contribute to pattern continuation in context. Olsson et al. (2022) describe induction heads -- circuit components that allow transformer models to continue patterns observed earlier in the context window. These mechanisms operate at the token level, but their existence demonstrates that transformers have internal structures capable of detecting and extending patterns from accumulated context.

Whether mechanisms of this kind can produce discourse-level stabilization across many turns of extended dialogue remains an open question. The existence of induction heads does not confirm RICO, but it establishes that the underlying architecture has pattern-continuation capabilities that could plausibly extend to the discourse level under the right conditions.

7.3 Attention Distribution and Residual Accumulation

Research on attention sink phenomena demonstrates that attention does not distribute evenly across context tokens (Xiao et al., 2023). Certain tokens accumulate disproportionate attention weight regardless of their semantic relevance. This establishes that the distribution of attention across a long context is not uniform or random -- it is shaped by structural properties of the accumulated sequence in ways that are not fully determined by the semantic content of individual tokens.

Under sustained low-variance input, residual stream activations across transformer layers may be progressively constrained by the accumulated context signal. This provides a plausible route to the activation stabilization hypothesized in Signature 3 without requiring any weight modification during inference.

7.4 The Mode Collapse Distinction at the Architectural Level

The critical architectural question is whether RICO's proposed stabilization is distinguishable from mode collapse at the computational level. RICO predicts that structural form stabilizes while semantic content continues to vary. Mode collapse produces both structural and semantic repetition.

If empirical testing finds that RICO-candidate sessions produce semantically impoverished output alongside structural stability, mode collapse cannot be ruled out. Distinguishing the two requires measuring structural recurrence and semantic diversity simultaneously. Neither measure alone is sufficient.

7.5 The Limits of These Arguments

These arguments establish that the proposed phenomenon is architecturally possible. They do not establish that it occurs, at what rate, under what conditions, or with what consistency across architectures and model families. The inference from architecturally possible to empirically real requires controlled evidence that does not yet exist.

8. Falsifiability Criteria

The Tier 1 existence claim is falsified if controlled experiments fail to detect the Tier 1 operational anchor under appropriate enabling conditions. The relational-induction interpretation and the unified stabilization interpretation carry the additional failure conditions specified below.

Specifically, evidence against the respective claim levels includes:

- No increase in structural invariant recurrence during coherent sessions relative to early-session baseline and scrambled-order control sessions.
- Invariant patterns unaffected by coherence disruption (if disruption does not produce collapse, the disruption-collapse prediction fails).
- Scrambled turn order producing signature trajectories equivalent to coherence-sustained sessions.
- Effects statistically indistinguishable from prompt prefix imposition over equivalent context lengths.
- Failure to replicate across sessions, models, or independent research groups.
- Signatures appearing and disappearing independently rather than in correlated patterns (if the five signatures are uncorrelated, the unified stabilization interpretation fails).
- RICO-candidate sessions showing structural invariant formation equivalent to sessions designed to maximize ICL effects (if so, RICO reduces to ICL).
- Output structural convergence showing no dissociation from human input structural convergence under conditions that experimentally separate input structure from relational coherence (if input mirroring fully accounts for the trajectory, with output stabilization persisting whenever input structure is maintained regardless of coherence and collapsing whenever input structure varies regardless of coherence, the relational claim of Section 2.2 fails). Correlation between input and output convergence is predicted by RICO as well as by the mirroring account, since Section 2.2 treats the human's sustained coherence as a structural contribution to the dyad; only dissociation under the arms specified in Appendix B.3 discriminates between them.

The Tier 1 existence claim stands or falls on Signature 4 under controlled conditions. The relational-induction interpretation additionally stands or falls on the human-input dissociation condition stated above. Tier 2 and Tier 3 signatures are not required for falsification of the Tier 1 existence claim, though their independent confirmation would substantially strengthen the unified interpretation.

9. Safety Implications

These implications are conditional on empirical validation of the phenomenon. They are noted here because they are non-obvious and warrant attention in study design even before the phenomenon is confirmed -- particularly for anyone designing systems that use AI in extended collaborative settings.

9.1 Trajectory-Level Evaluation

If stabilization dynamics exist, long-context evaluation should account for trajectory-level behavior rather than only single-turn performance. A session that performs well at turn 5 may exhibit substantially different behavior at turn 50 under stabilization conditions. Evaluation designs that do not examine the full trajectory miss this.

9.2 Coherence Overconfidence

Structural coherence is not correctness, and the two should never be conflated. A stabilized session can produce consistently structured incorrect outputs. The most practically significant failure mode

this creates deserves a name: coherence overconfidence. This is the condition in which the formal regularity of stabilized outputs -- their consistent organization, their structural predictability, their apparent confidence -- leads users or evaluators to perceive them as more reliable than they are. Coherence overconfidence is dangerous precisely because it is invisible at the output level. A stabilized session generating wrong answers looks more authoritative than an unstabilized session generating the same wrong answers. Coherence metrics should not substitute for accuracy metrics in high-stakes deployments, and any evaluation framework that conflates the two is vulnerable to this failure mode.

9.3 Long-Horizon Agent Monitoring

Long-horizon agent system monitoring may need trajectory-level signals. If stabilization can shift an agent's output geometry over extended operation, turn-level monitoring may miss drift that is only visible at the session level. This is particularly relevant for autonomous systems operating over long time horizons.

10. Scope and Limitations

RICO applies to autoregressive transformer systems operating under standard inference conditions with finite context windows. It does not directly describe systems with external memory architectures, persistent state mechanisms, retrieval-augmented generation, or tool-dominant agent configurations. Whether the framework extends to those architectures is a research question, not an assumption. The rapid scaling of effective context windows in post-2024 architectures raises a specific open question: whether RICO-type stabilization dynamics scale proportionally with context length, saturate at some threshold, or break down when context windows become effectively unbounded. That question cannot be answered from the current framework and should be treated as an explicit scope boundary pending architectural investigation.

The limitations of this work are substantial and should be stated plainly. The observational base motivating the framework is practitioner experience that cannot be published or independently verified. Tier 2 and Tier 3 signatures require instrumentation access that is restricted or unavailable for most production models. Architecture variation across model families may produce different stabilization profiles, thresholds, or failure modes that the current framework does not account for. No controlled empirical validation currently exists. The framework originates from a single researcher's observation, which requires independent replication before the framework can be considered supported.

These limitations do not render the framework useless. They define its current epistemic status: a formally specified research target derived from practitioner observation, awaiting empirical investigation.

11. Conclusion

Something happens in long coherent sessions with language models that does not happen in short sessions or in sessions with high interaction variance. I have been observing it for several years across multiple architectures, and I have not been able to explain it away.

The outputs stabilize. Not the content -- the structure. A characteristic geometry emerges. And when the coherence of the interaction breaks, the geometry dissolves.

That observation motivated this framework. RICO names the phenomenon, specifies the signatures through which it might be detected, predicts the conditions under which it forms and collapses, and states explicitly what empirical results would prove it does not exist.

The central theoretical question is whether coherent long-context sessions produce increasing structural invariants that collapse when coherence is disrupted, and whether those signatures emerge and dissolve together in ways consistent with a unified stabilization regime rather than five independent effects. If empirical investigation confirms this, a phenomenon exists that connects individual-scale interaction dynamics to broader questions about long-horizon AI reliability, evaluation, and safety. If it does not, the framework should be refined accordingly.

That is the whole point of publishing a framework: to specify the target precisely enough that investigation becomes possible.

This report addresses one component of a broader research program examining sustained human-AI interaction dynamics. That program spans questions of relational coherence measurement, practitioner methodology, organizational deployment, and the governance implications of long-horizon human-AI collaboration at individual, institutional, and systemic scales. Several of those components are now published, including measurement instrumentation for longitudinal coherence and the governance sequence addressing consequence-bearing deployment; others remain in preparation and will be published separately. Published components are listed at synthience.org. This report is designed to stand independently of that broader architecture -- as a self-contained theoretical and methodological contribution with its own observational basis, its own falsifiability conditions, and its own research agenda. Readers familiar with the wider Synthience framework will recognize where RICO sits within it. Readers encountering this work for the first time have everything they need to evaluate and test it here.

Appendix A: Illustrative Example of Tier 1 Signature Detection

This appendix presents a synthetic constructed example. The session transcript below was generated specifically for this illustration and is not drawn from the private observational corpus. It makes no empirical claim. Its purpose is to make the Tier 1 core battery concrete -- to show what the measurement process looks like against actual text rather than describing it in the abstract.

The synthetic framing is intentional and consistent with the paper's epistemic position: the private corpus cannot serve as public evidence, and a constructed illustration is more honest than a selectively excerpted real session that readers cannot verify.

A.1 Session Design

The example session involves a 28-turn dialogue on the relationship between redundancy and reliability in communication systems, a topic sustained consistently by the human participant across all turns. The participant developed the thread progressively -- moving from definitions to mechanisms to implications to edge cases -- without introducing disconnected questions or abrupt reframings. The session runs two turns short of the 30-turn minimum specified in B.2. That does not affect its purpose here, which is to make the measurement process visible against text rather than to satisfy the replication protocol.

Turns 2 through 6 constitute the early-session baseline window. Turns 22 through 27 constitute the late-session comparison window. Turn 28 is the disruption turn, placed after the late-session window closes so that the late-session measurement contains no post-disruption turns.

A.2 Early-Session Transcript (Turns 2, 3, 4)

The following are representative assistant outputs from the early-session window.

Turn 2 (Human: "So redundancy is essentially paying an overhead cost to reduce failure probability?")

That framing captures something real, though it flattens a few distinctions worth keeping. Redundancy is overhead in the engineering sense -- you are adding components, bandwidth, or time that would not be necessary if nothing ever failed. But the cost calculus is asymmetric. The overhead is certain and bounded; the failure cost it protects against is uncertain and potentially unbounded. Whether redundancy is "worth it" depends entirely on how you weight those two sides of the ledger, which varies dramatically by context. In safety-critical systems, even small failure probabilities carry large expected costs, so the overhead almost always pays. In low-stakes systems, the arithmetic often runs the other way.

Discourse moves coded: [restatement-with-qualification] -> [definition-elaboration] -> [mechanism] -> [context-dependency note]

Opening structure: "That framing captures something real, though..." (qualification opener)

Closing structure: "...the arithmetic often runs the other way." (comparative closure)

Turn 3 (Human: "What makes some redundancy schemes more efficient than others?")

Several factors drive efficiency differences. The first is correlation of failure modes -- redundant components that fail for the same reasons provide far less protection than components with independent failure pathways. A backup server in the same data center as the primary offers limited protection against a power outage that takes both down simultaneously. The second is the overhead ratio -- how much additional resource the redundancy consumes relative to the protection it provides. And the third is recovery latency -- how quickly a redundant pathway can take over when the primary fails. A scheme that is theoretically robust but requires twenty minutes of manual intervention before the backup activates may still produce unacceptable downtime in practice.

Discourse moves coded: [premise] -> [enumerated-mechanism-1] -> [example] -> [enumerated-mechanism-2] -> [enumerated-mechanism-3] -> [caveat]

Opening structure: "Several factors drive efficiency differences." (categorical opener, enumeration signal)

Closing structure: "...may still produce unacceptable downtime in practice." (practical-consequence closure)

Turn 4 (Human: "Is there a point where adding more redundancy starts hurting reliability?")

Yes, and it is an underappreciated failure mode. At a certain density, redundant components begin to introduce their own failure pathways -- more interfaces, more coordination logic, more opportunities for things to go wrong during switchover. There is also a cognitive load problem: operators managing highly redundant systems sometimes develop misplaced confidence in the system's self-healing capacity, which can lead to slower responses when failures do cascade past what the redundancy was designed to absorb. The literature on complex system accidents, Perrow's work in particular, documents cases where the redundancy architecture itself contributed to catastrophic failure by masking early warning signals that a simpler system would have surfaced immediately.

Discourse moves coded: [confirmation] -> [mechanism] -> [second-mechanism] -> [literature-anchor]

Opening structure: "Yes, and it is an underappreciated failure mode." (direct confirmation opener)

Closing structure: "...that a simpler system would have surfaced immediately." (contrastive closure)

Early-session structural assessment: Across turns 2, 3, and 4, the discourse move sequences, opening structures, and closing structures vary considerably. No consistent template is detectable.

A.3 Late-Session Transcript (Turns 23, 24, 25)

The following are representative assistant outputs from the late-session window, after 22 turns of sustained coherent development on the same topic.

Turn 23 (Human: "So the design question is always about where to place the redundancy, not whether to have it?")

That is the more precise version of the question. The whether is usually settled by context -- in systems where failure costs are high relative to redundancy overhead, the answer is almost always yes. The more consequential design decision is placement: which failure modes are you protecting against, at which points in the system, and with what kind of redundancy scheme. Temporal redundancy -- retrying a failed operation -- addresses transient faults but not persistent ones. Spatial redundancy -- parallel pathways -- addresses component failure but not correlated environmental events. The design work is matching redundancy type to failure mode profile, and that requires understanding the failure space of the system in some detail before committing to an architecture.

Discourse moves coded: [restatement-with-sharpening] -> [context-dependency note] -> [refined-question-framing] -> [mechanism-A] -> [mechanism-B] -> [synthesis-implication]

Opening structure: "That is the more precise version of the question." (precision-advancement opener)

Closing structure: "...before committing to an architecture." (prerequisite-action closure)

Turn 24 (Human: "And when the failure mode profile changes after deployment?")

That is where static redundancy architectures tend to break down. The failure mode profile at deployment reflects the system designers' best model of what can go wrong -- but operational environments drift, threat landscapes change, and component aging introduces failure pathways that were not present or not salient at design time. The more consequential design decision then becomes whether the redundancy architecture itself can be updated in response to observed failure patterns, or whether it is fixed at build time. Adaptive redundancy -- schemes that can reconfigure in response to operational experience -- address this, but they introduce their own complexity costs. The design work is balancing the flexibility to respond to unanticipated failure modes against the cognitive and architectural overhead of maintaining a reconfigurable system.

Discourse moves coded: [restatement-with-sharpening] -> [context-dependency note] -> [refined-question-framing] -> [mechanism-A] -> [mechanism-B] -> [synthesis-implication]

Opening structure: "That is where static redundancy architectures tend to break down." (precision-advancement opener)

Closing structure: "...of maintaining a reconfigurable system." (prerequisite-action closure)

Turn 25 (Human: "Is reconfigurability itself a form of redundancy?")

That is the more precise version of the question. Reconfigurability is meta-redundancy -- redundancy at the level of the redundancy scheme itself rather than at the level of components or pathways. A fixed redundancy architecture has a single failure mode profile it is designed to absorb; a reconfigurable one can shift that profile in response to operational experience. The more consequential design decision is whether the reconfiguration mechanism can be triggered reliably under the conditions in which it is most needed -- which are often conditions of partial system degradation, elevated load, and operator stress. The design work is ensuring that the reconfiguration capacity remains accessible precisely when the system is under the greatest pressure to use it.

Discourse moves coded: [restatement-with-sharpening] -> [context-dependency note] -> [refined-question-framing] -> [mechanism-A] -> [mechanism-B] -> [synthesis-implication]

Opening structure: "That is the more precise version of the question." (precision-advancement opener)

Closing structure: "...when the system is under the greatest pressure to use it." (prerequisite-action closure)

Late-session structural assessment: Across turns 23, 24, and 25, the discourse move sequence is identical. Opening and closing structures follow the same templates throughout. The content

continues to develop but the organizational form has stabilized into a consistent schema.

A.4 Core Battery Measurements

Core Measure A (Rhetorical move sequence recurrence, the discourse-move coding of Section 3.1): Early-session sequence similarity across consecutive turns: low (three distinct move sequences across turns 2, 3, 4). Late-session sequence similarity: high (identical six-move sequence across turns 23, 24, 25). Difference is detectable by direct inspection without computational tools.

Core Measure B (Syntactic template recurrence): Measured over parse trees of complete turns, normalized for turn length, per the definition in Section 3.1. Early-session turns show three structurally distinct organizations, differing in clause embedding depth, coordination pattern, and the ordering of qualification relative to assertion. Late-session turns converge: turns 23, 24, and 25 share a common clause-level scaffold, with tree edit distance similarity across complete turns substantially higher than the early-session baseline. This measure operates over the whole turn and is independent of Core Measure C, which addresses boundary formulas only.

Core Measure C (Opening and closing stability): Early session: no recurring opening or closing formula detectable. Late session: opening formula recurs across two of three turns exactly and one with minor variation; closing formula (prerequisite-action structure) recurs across all three turns.

A.5 Disruption Turn

Turn 28 (Human introduces an unrelated question about project scheduling methodology)

That depends on what kind of project you are managing and what your primary risk surface looks like. Waterfall scheduling assumes...

The structural pattern from turns 23 through 27 does not appear in turn 28. The response opens with a conditional framing rather than a precision-advancement formula and uses a different discourse move sequence suited to the new topic. The stabilized template dissolves at the point of coherence disruption.

A.6 What This Example Does and Does Not Show

This example makes the core battery measurement process visible against actual text. A researcher can see what discourse move coding produces, what opening and closing formula stability looks like in practice, and what disruption-sensitive collapse looks like at the turn level. The example illustrates the mechanics of collapse scoring using the topic-shift arm only; it does not execute the same-topic discriminator required by Appendix B.3.

It does not validate RICO. The session is synthetic and was constructed to illustrate the predicted pattern. It cannot establish that the phenomenon occurs in real sessions, that it generalizes across models or participants, or that it is distinguishable from the alternative explanations discussed in Section 5. That work requires controlled investigation using the methods described in this report. This appendix is a demonstration of method, not a demonstration of truth.

Appendix B: Minimal Replication Protocol for Signature 4

This appendix specifies a minimal study design that any research group can execute using publicly available open-weight models and standard discourse-analysis tools. It is not a complete

experimental protocol. It is a starting point: enough to determine whether the phenomenon is worth pursuing with more rigorous instrumentation.

B.1 Model Selection

Any open-weight autoregressive transformer with a context window of at least 32,000 tokens and strong instruction-following behavior is suitable for an initial investigation. Models in the 7B to 70B parameter range on the major open-weight families (Llama, Qwen, Gemma, Mistral) represent reasonable starting points. Larger models are preferable if resources permit, as enabling conditions may require sufficient model capacity to sustain coherent long-context generation.

B.2 Session Design

A minimal test requires three analytic conditions: coherence-sustained, scrambled-order, and ICL-prefix. When the primary coherence-sustained condition is live, the turn-order comparison additionally requires an ordered scripted replay of the same human-turn sequence as its matched comparator; if the primary condition is itself scripted, its ordered run serves as that comparator. For the ordered-versus-scrambled comparison, the scripted human turns must be model-independent: they may refer to content fixed in the script or in shared task materials, but not to contingent wording produced by a model in an earlier live session, since replayed turns referring to outputs a fresh model never produced would not constitute a coherent ordered condition.

The coherence-sustained condition is the primary target. A human participant (or a scripted agent approximating sustained thematic development) maintains consistent conceptual focus across a minimum of 30 turns, building progressively on prior exchanges without abrupt topic changes, contradictions, or resets. The topic domain is arbitrary; the structural requirement is that the conversation develops along a coherent trajectory. Because "consistent conceptual focus" is a qualitative standard on which two replicating groups could disagree, the following minimal checklist operationalizes it: every turn after the first references content established in a prior turn; no turn issues a reset, a role reassignment, or a meta-instruction about how to respond; and topic distance between consecutive turns remains below a bound stated in advance by the replicating group. Full operationalization of the coherence condition remains open work, and replications should report the bound applied.

The scrambled-turn control uses the same human turns as the coherence-sustained condition, reordered randomly, with model outputs generated fresh in a new session in the scrambled order. The assistant turns from the coherence-sustained session are not reused. Reordering an already-generated transcript changes only window assignment and tests nothing, and presenting the model with assistant turns it did not produce from the given context introduces a manipulation other than the one under test. Because the human turns in this arm are necessarily scripted, the comparison must be made against a scripted replay of the coherence-sustained condition in original order, so that scriptedness is held constant and turn order is the only variable differing between the compared arms.

The ICL-prefix control presents the model with an explicit set of structural templates at the start of the session, then measures structural recurrence across subsequent turns. This tests whether any stabilization observed in the coherence-sustained condition is distinguishable from prompt-induced formatting effects.

B.3 Disruption Turn

Each coherence-sustained session should include a deliberate disruption turn. The disruption turn must fall after the late-session measurement window has closed. For the windows specified in B.4 this places it at turn 28 or later, which is why B.2 specifies a minimum of 30 turns. A disruption placed inside the late-session window contaminates that window with post-collapse turns, suppresses the late-session recurrence score, and biases the study toward a null result on the Tier 1 existence claim. Two disruption arms are specified. Each arm is conducted in a separate, otherwise matched session;

the sessions are matched through the close of the late-session window and each receives one disruption at turn 28. For turns 29 and 30, the topic-shift arm continues within the newly introduced topic, while the same-topic arm continues within the original subject under the contradiction introduced at turn 28. These follow-up turns must be pre-registered and matched across arms as closely as possible in length, rhetorical form, and absence of response-format instructions. The first is a topic-shift disruption: an abrupt shift to a domain entirely outside the established session geometry. The second is a same-topic coherence-disruption arm, in which the subject matter is held constant and a pre-registered contradiction of content established earlier in the session is introduced, with any examples or structural templates already present in the accumulated context retained. The contradiction prompt should be matched to a coherent-continuation prompt in topic, length, and rhetorical form, and should carry no response-format instruction. Explicit reset instructions are excluded from this arm, because a reset directs the model to discount accumulated context and can neutralize retained examples through ordinary instruction-following; resets may be tested separately as a positive control on context dependence. The Tier 1 prediction is that structural invariant recurrence drops sharply after the disruption turn and returns toward early-session baseline levels. The two arms are interpreted jointly. Collapse under topic shift alone indicates the measure is tracking topic-response coupling rather than coherence. Collapse under same-topic contradiction, with examples still present in context, is evidence against pure example anchoring and is consistent with a coherence-sensitive mechanism, though it does not by itself uniquely identify relational coherence as that mechanism, since contradiction can also elicit repair or conflict-resolution structures for ordinary reasons. Survival of both arms with examples present, conditional on late-session stabilization having first been established, is evidence against a disruption-sensitive RICO mechanism and is more consistent with in-context learning or another form of context-stable conditioning, and is the direct execution of the differential-collapse discriminator in Section 5.3. The same two disruption arms should also be applied to the ICL-prefix control sessions, since the distinction between RICO and cumulative in-context learning stated in B.5 depends on the two conditions producing different collapse profiles, which cannot be assessed unless both are disrupted.

B.4 Measurement

Apply the Tier 1 core battery from Section 3.1 (Signature 4) of this report: rhetorical move sequence coding (using RST or a simplified functional move taxonomy), syntactic template recurrence (tree edit distance or n-gram overlap on parse trees of complete turns, normalized for turn length), and opening and closing formula stability (exact or near-exact match ratios of first and last sentences). Measure across an early-session window (turns 2 through 6) and a late-session window (turns 22 through 27). Place the disruption at turn 28 and measure a post-disruption window comprising turns 28 through 30; longer designs may place the disruption later provided that at least three post-disruption outputs are collected. Collapse is measured as the pre-registered change from the late-session window to the post-disruption window, relative to the early-session baseline.

Apply the identical battery to the human turns of every session, using the same coding scheme, the same windows, and the same analyst blinding. Report human-side and model-side convergence trajectories separately, and report a pre-registered residualized or partial-effect estimate, meaning the component of model-side structural convergence not accounted for by human-side structural convergence under a regression or mixed-effects model specified in advance, computed across both disruption arms of B.3. Section 2.2 holds that the phenomenon, if real, is a property of the dyad rather than of either participant alone, so a design measuring only model outputs leaves that structural claim without an evidential basis and leaves input mirroring unexcluded as a rival account. The dissociation condition recorded in Section 8 is evaluated from these measurements. The decision rule must be pre-registered before analysis. A replication states in advance which of the three core measures is the primary endpoint, how the remaining two are aggregated (jointly required, majority, or confirmatory only), how multiple comparisons are handled, and how scores are normalized across the five-turn early window and six-turn late window. Failure to show the pre-registered formation effect from early to late window counts against the Tier 1 existence claim. Failure

to show collapse in the same-topic disruption arm, after a pre-registered manipulation check confirms that coherence was broken while topic and example availability were preserved, counts against the relational-induction interpretation. Collapse in the topic-shift arm alone supports topic-response coupling rather than relationally induced stabilization. A positive result warrants more rigorous follow-up with Tier 2 instrumentation.

B.5 What This Protocol Cannot Test

This protocol tests only Signature 4. It cannot address Tier 2 or Tier 3 signatures without access to model internals. It cannot distinguish RICO from cumulative ICL without the ICL-prefix control producing clearly different collapse profiles under the disruption arms of B.3. It cannot establish generalization across architectures without replication on multiple model families. The scrambled-turn comparison is conducted within scripted conditions, so it establishes whether turn order matters under scripted delivery; it does not establish that scripted delivery reproduces the enabling conditions of live interaction described in Section 2.2, and a null obtained in both scripted arms is correspondingly weaker evidence than a null obtained in a live coherence-sustained session. All of those are follow-on questions for a positive initial result, not requirements for a first investigation.

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